

# Cheat Sheet: Pandas, Matplotlib & Seaborn

You don't have to remember syntax - you have to know WHAT you want to do. Find it here.

## PANDAS - working with tables

### Setup & loading

```
import pandas as pd
import numpy as np
df = pd.read_csv('file.csv')
```

### Look at the data

```
df.head()          # first 5 rows
df.tail(3)         # last 3 rows
df.shape           # (rows, columns)
df.columns         # column names
df.info()          # types + missing
df.describe()      # stats of numbers
df.dtypes          # type of each column
```

### Pick columns / rows

```
df['Rating']        # one column
df[['Title', 'Price']] # many columns
df.iloc[0]          # row by position
df.iloc[0:5]        # rows 0-4
df.loc[3, 'Title']   # row 3, col Title
```

### Filter rows (conditions)

```
df[df['Price'] < 20]
df[df['Genre'] == 'RPG']
df[df['Multiplayer'] != 'No']
# combine: & = and, | = or, ~ = not
df[(df['Price'] < 20) & (df['Rating'] > 8)]
df[(df['Genre'] == 'RPG') | (df['Genre'] == 'Puzzle')]
df[df['Genre'].isin(['RPG', 'Puzzle'])]
df[df['Title'].str.contains('Star')]
```

### Group by (split -> calculate -> combine)

```
df.groupby('Genre')['Price'].mean()
df.groupby('Genre')['Price'].max()
df.groupby('Genre').size()          # count rows
df.groupby('Platform')['Sold'].sum()
df.groupby(['Genre', 'Platform'])['Price'].mean()
df.groupby('Genre')['Price'].agg(['mean', 'max'])
# sort the result
(df.groupby('Genre')['Price'].mean()
 .sort_values(ascending=False))
```

### Quick math on a column

```
df['Rating'].mean()      # average
df['Rating'].median()    # middle value
df['Rating'].max()       # biggest
df['Rating'].min()       # smallest
df['Rating'].sum()       # non-missing
df['Rating'].count()     # spread
df['Rating'].std()       # different values
df['Genre'].unique()     # how many different
df['Genre'].nunique()    # count each
df['Genre'].value_counts()
```

### Find the row with max / min

```
df['Rating'].idxmax()    # index of max
df.loc[df['Rating'].idxmax()] # the whole row
df.loc[df['Rating'].idxmax(), 'Title']
```

### Sort

```
df.sort_values('Price')          # low -> high
df.sort_values('Price', ascending=False) # high -> low
df.sort_values(['Genre', 'Price']) # by two
df.nlargest(5, 'Rating')         # top 5
df.nsmallest(3, 'Price')         # bottom 3
```

### Make new columns

```
df['Total'] = df['Price'] * df['Copies']
df['Age'] = 2026 - df['Release_Year']
df['Is_Free'] = df['Price'] == 0
df['Cheap'] = np.where(df['Price'] < 20, 'Yes', 'No')
df = df.drop(columns=['Total']) # remove
df = df.rename(columns={'old': 'new'})
```

### Missing values (NaN)

```
df.isna().sum()          # how many per column
df[df['Rating'].isna()]   # rows with missing
df.dropna()              # remove rows w/ missing
df['Rating'].fillna(0)    # fill with 0
df['Rating'].fillna(df['Rating'].mean())
df['Rating'] = df['Rating'].fillna(
    df['Rating'].median())
```

### Useful extras

```
df.corr(numeric_only=True) # correlation table
df.reset_index(drop=True)  # renumber rows
df.copy()                  # safe copy
len(df)                    # number of rows
```

# MATPLOTLIB - drawing plots

## Setup & the basic recipe

```
import matplotlib.pyplot as plt

plt.figure(figsize=(8, 5)) # width, height
# ... draw something ...
plt.title('My Title')
plt.xlabel('X label')
plt.ylabel('Y label')
plt.show()
```

## Plot types

```
plt.plot(x, y) # line
plt.plot(x, y, marker='o') # line w/ dots
plt.bar(names, values) # vertical bars
plt.barh(names, values) # horizontal bars
plt.scatter(x, y) # dots
plt.hist(values, bins=10) # histogram
plt.pie(values, labels=names) # pie
plt.boxplot(values) # box plot
```

## Plot straight from pandas

```
df['Price'].plot(kind='hist')
df['Genre'].value_counts().plot(kind='bar')
df.plot(kind='scatter', x='Price', y='Rating')
df.groupby('Genre')['Price'].mean().plot(kind='bar')
```

# SEABORN - prettier statistical plots

## Setup

```
import seaborn as sns
sns.set_theme() # nice default style
# seaborn plots still use plt.title(), plt.show()
```

## Counting categories

```
sns.countplot(data=df, x='Genre')
sns.countplot(data=df, x='Platform',
              hue='Multiplayer')
```

## Comparing groups

```
sns.barplot(data=df, x='Genre', y='Price') # avg
sns.boxplot(data=df, x='Genre', y='Rating')
sns.violinplot(data=df, x='Genre', y='Rating')
sns.stripplot(data=df, x='Genre', y='Rating')
```

## Make it look nice

```
plt.xticks(rotation=45) # tilt labels
plt.legend() # show legend
plt.grid(True) # grid lines
plt.tight_layout() # fix overlaps
plt.plot(x, y, color='red', label='Sales')
plt.scatter(x, y, s=50, alpha=0.6) # size, fade
plt.ylim(0, 10) # y-axis range
```

## Several plots in one figure

```
fig, axes = plt.subplots(1, 2, figsize=(12, 4))
axes[0].hist(df['Rating'])
axes[0].set_title('Ratings')
axes[0].set_xlabel('Rating')
axes[0].set_ylabel('Count')
axes[1].scatter(df['Price'], df['Rating'])
axes[1].set_title('Price vs Rating')
plt.tight_layout()
plt.show()
# NOTE: on an axis use set_title / set_xlabel
# instead of plt.title / plt.xlabel
```

## Numbers vs numbers

```
sns.scatterplot(data=df, x='Price', y='Rating')
sns.scatterplot(data=df, x='Price', y='Rating',
               hue='Genre', size='Copies')
sns.lineplot(data=df, x='Year', y='Price')
sns.regplot(data=df, x='Price', y='Rating') # +trend
```

## Distributions

```
sns.histplot(data=df, x='Rating', bins=10)
sns.histplot(data=df, x='Rating',
              hue='Multiplayer')
sns.kdeplot(data=df, x='Rating')
```

## Heatmap & pairs

```
corr = df.corr(numeric_only=True)
sns.heatmap(corr, annot=True, cmap='coolwarm')
sns.pairplot(df[['Price', 'Rating', 'Copies']])
```

# WHICH PLOT SHOULD I USE?

| I want to...                              | Use this               | Example                |
|-------------------------------------------|------------------------|------------------------|
| Count how many in each category           | bar / countplot        | games per Genre        |
| Compare an average across categories      | bar / barplot          | avg Price per Platform |
| See how one number is spread out          | hist / histplot        | distribution of Rating |
| Compare spread across groups (+ outliers) | boxplot                | Rating by Genre        |
| See if two numbers are related            | scatter / scatterplot  | Price vs Copies sold   |
| Show change over time                     | plot (line) / lineplot | games per Year         |

| I want to...                 | Use this                          | Example                     |
|------------------------------|-----------------------------------|-----------------------------|
| See ALL correlations at once | heatmap of <code>df.corr()</code> | which columns move together |
| Show parts of a whole        | pie                               | share of each Platform      |

**Stuck? Ask yourself:** 1) Which column(s) do I need? 2) Do I filter rows first? 3) Do I group? 4) What one number or plot answers the question? Then look up the syntax above. **Read error messages** - the last line usually tells you exactly what's wrong.